

Commonly Used Distribution – I

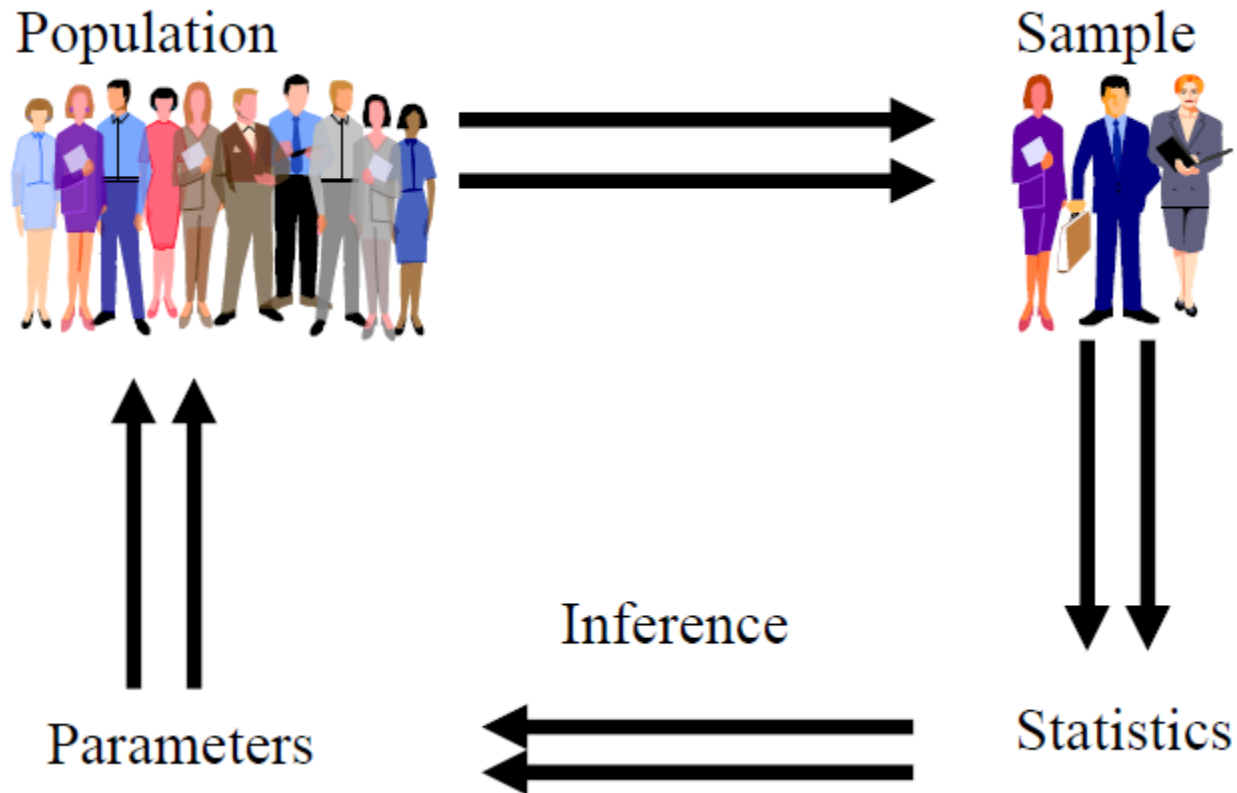
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References:

- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, Chapter 3
- Walpole R. E., Myers R. H., Myers S. L. and Ye K., *Probability & Statistics for Engineers & Scientists*, Ch. 5
- William Navidi, *Statistics for Engineers and Scientists*, Chapter 4
- Slides are credited from Prof. Berlin Chen, NTNU.

How to Estimate Population (Distribution) Parameters?



- Unbiased Estimate ?
- Estimation Uncertainty ?

Propagation Error

- Any measuring procedure contains error
 - This causes measured values to differ from the true values that are measured
- Errors in measurement produce error in calculated values (like the mean)
- Definition: When error in measurement produces error in calculated values, we say that error is **propagated** from the measurements to the calculated value
- Having knowledge concerning the sizes of the errors in measurement → Obtaining knowledge concerning the likely size of the error in a calculated quantity

Measurement Error

- A geologist weighs a rock on a scale and gets the following measurements:
251.3 252.5 250.8 251.1 250.4
- None of the measurements are the same and none are probably the **actual** measurement
- The **error** in the measured value is the difference between a measured value and the true value

Parts of Error

- We think of the error of the measurement as being composed of two parts:
 - Systematic error (or bias)
 - Random error
- Bias is error that is the same for every measurement
 - E.g., a **imperfectly calibrated scale** always gives you a reading that is too low
- Random error is error that varies from measurement to measurement and averages out to zero in the long run
 - E.g., Parallax (視差) error $E[\text{Random Error}] = 0$
 - The difference in the position of dial indicator when observed from different angles

$$\begin{array}{ccccccc} \text{Measured Value} & = & \text{True Value} & + & \text{Bias} & + & \text{Random Error} \\ \text{random variable} & & \text{constant} & & \text{constant} & & \text{random variable} \end{array}$$

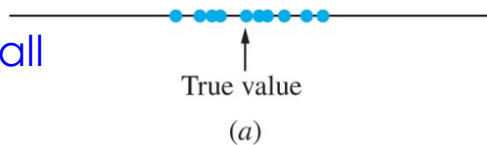
Two Aspects of the Measuring Process (1/4)

- We are interested in **accuracy**
 - Accuracy is determined by bias
 - The bias in the measuring process is **the difference between the mean measurement μ and the true value**:
$$\text{bias} = \mu - \text{true value}$$
 - The smaller the bias, the more accurate the measuring process
 - **Unbiased**: the mean measurement is equal to the true value
- The other aspect is **precision**
 - **Precision refers to the degree to which repeated measurements of the same quantity tend to agree with each other**
 - If repeated measurements come out nearly the same every time, the precision is high
 - The **uncertainty** in the measuring process is the standard deviation σ
 - The smaller the uncertainty, the more precise the measuring process

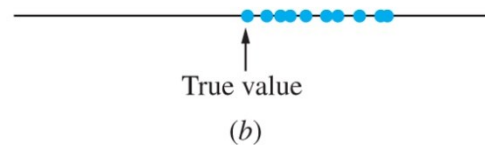
Two Aspects of the Measuring Process (2/4)

- Example: Figures 3.1 and 3.2

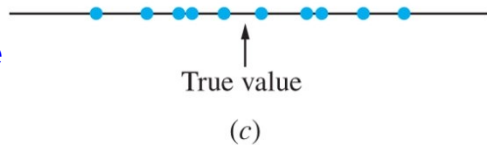
both bias and
uncertainty are small



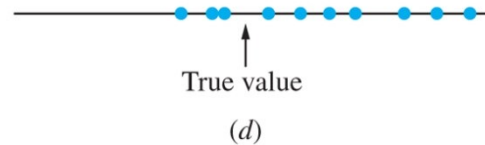
bias is large;
uncertainty is small



bias is small;
uncertainty is large

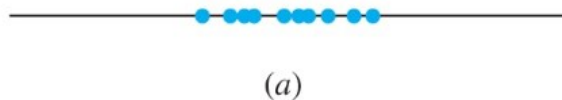


both bias and
uncertainty are large

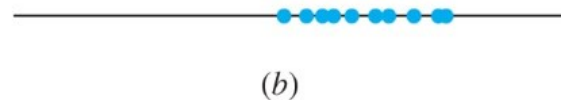


How about bias? (True value is often unknown in real life)

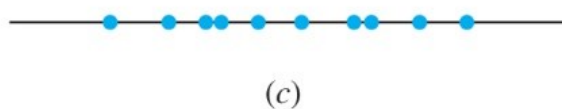
uncertainty
is small



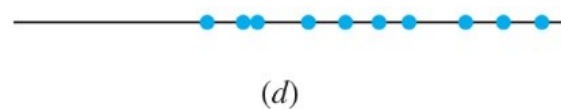
uncertainty
is small



uncertainty
is large



uncertainty
is large



Two Aspects of the Measuring Process (3/4)

- Let $X_1, \dots, X_n$ be independent measurements, all made by the same process on the same quantity
 - The **sample standard deviation s** can be used to **estimate the uncertainty**
 - Estimates of uncertainty are often crude, especially when based on small samples
 - If the true value is known, the **sample mean, $\bar{X}$** , can be used to **estimate the bias**:

$$\text{bias} \approx \bar{X} - \text{true value}$$

- If the true value is unknown, the bias cannot be estimated from repeated measurements

Two Aspects of the Measuring Process (4/4)

- From now on, we will describe measurements in the form

$$\text{Measured value} \pm \sigma$$

- Where σ is the **uncertainty** in the process that produced the measured value
- Assume that bias has reduced to a negligible level

Linear Combinations of Measurements

- Each measurement can be viewed as a random variable
 - Its **standard deviation** represents the **uncertainty** for it
- How to compute uncertainties in scaled measurements or combinations of **independent** measurements?
 - If X is a measurement and c is a constant, then

$$\sigma_{cX} = |c| \sigma_X$$

- If $X_1, \dots, X_n$ are **independent** measurements and $c_1, \dots, c_n$ are constants, then

$$\sigma_{c_1X_1 + \dots + c_nX_n} = \sqrt{c_1^2 \sigma_{X_1}^2 + \dots + c_n^2 \sigma_{X_n}^2}$$

Example

- Example 3.6: A surveyor is measuring the perimeter of a rectangular lot. He measures two adjacent sides to be $50.11 \pm 0.05\text{m}$ and $75.21 \pm 0.08\text{m}$. These measurements are independent. Estimate the perimeter of the lot and find the uncertainty in the estimate.

– Solution:

Let $X = 50.11$ and $Y = 75.21$ be the two measurements. The perimeter is estimated by $P = 2X + 2Y = 250.64\text{m}$, and the uncertainty in P is

$$\sigma_P = \sigma_{2X+2Y} = \sqrt{4\sigma_X^2 + 4\sigma_Y^2} = \sqrt{4(0.05)^2 + 4(0.08)^2} = 0.19\text{m}$$

So the perimeter is $250.64 \pm 0.19\text{m}$.

Repeated Measurements

- One of the best ways to reduce uncertainty is to take independent measurements and average them (**why?**)
 - The measurements can be viewed as a simple random sample from a population
 - Their average is therefore the sample mean
- If $X_1, \dots, X_n$ are **independent** measurements, each with mean μ and standard deviation σ , then the **sample mean**, $\bar{X}$, is a measurement with mean (**cf. Sections 2.5 & 2.6**)

$$\mu_{\bar{X}} = \mu$$

and with **uncertainty**

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

The uncertainty is reduced by a factor equal to the square root of the sample size

Example

- Example 3.8: A length of a component is to be measured by a process whose uncertainty is 0.05 cm. If 25 independent measurements are made and the average of these is used to estimate the length, what will the uncertainty be? How much more precise is the average of 25 measurements than a single measurement?

– Solution:

The uncertainty is $0.05 / \sqrt{25} = 0.01$ cm. The average of 25 independent measurements is five times more precise than a single measurement.

Sample Mean and Variance

- *Point estimation* is the attempt to estimate some fixed but unknown quantity θ of a random variable by using sample data
- Let $\mathbb{X} = \{x^{(1)}, \dots, x^{(n)}\}$ be a set of n i.i.d. samples of a random variable X , a *point estimator* or *statistic* is a function of the data:

$$\hat{\theta}_n = g(x^{(1)}, \dots, x^{(n)})$$

- The value $\hat{\theta}_n$ is called *estimate* of θ
- Sample mean: $\bar{X} = \hat{\mu}_X = \frac{1}{n} \sum_i x^{(i)}$
- Sample variance: $s^2 = \hat{\sigma}_X^2 = \frac{1}{n} \sum_i (x^{(i)} - \hat{\mu}_X)^2$
- How good are these estimators?

Bias & Variance

- *Bias* of an estimator

$$\text{bias}(\hat{\theta}_n) = E_{\mathbb{X}}(\hat{\theta}_n) - \theta$$

- Here, the expectation is defined over *all possible $\mathbb{X}$'s of size n* , i.e.,
 $E_{\mathbb{X}}(\hat{\theta}_n) = \int \hat{\theta}_n dP(\mathbb{X})$

- We call a statistic *unbiased estimator* iff it has zero bias
- *Variance* of an estimator:

$$\text{Var}_{\mathbb{X}}(\hat{\theta}_n) = E_{\mathbb{X}} \left[(\hat{\theta}_n - E_{\mathbb{X}}[\hat{\theta}_n])^2 \right]$$

- Is $\hat{\mu}_X = \frac{1}{n} \sum_i x^{(i)}$ an unbiased estimator of μ_X (population mean) ? **Yes**
- What much is $\text{Var}_{\mathbb{X}}(\hat{\mu}_X)$

from Prof. Wu, Shan-Hung's slide

Variance of $\hat{\mu}_X$

$$\begin{aligned}\text{Var}_{\mathbb{X}}(\hat{\mu}_X) &= E_{\mathbb{X}}[(\hat{\mu} - E_{\mathbb{X}}[\hat{\mu}])^2] = E[\hat{\mu}^2] - \mu^2 \\&= E\left[\frac{1}{n^2} \sum_{i,j} x^{(i)} x^{(j)}\right] - \mu^2 = \frac{1}{n^2} \sum_{i,j} E[x^{(i)} x^{(j)}] - \mu^2 \\&= \frac{1}{n^2} \left(\sum_{i=j} E[x^{(i)} x^{(j)}] + \sum_{i \neq j} E[x^{(i)} x^{(j)}] \right) - \mu^2 \\&= \frac{1}{n^2} \left(\sum_i E[x^{(i)2}] + n(n-1)E[x^{(i)}]E[x^{(j)}] \right) - \mu^2 \\&= \frac{1}{n} E[X^2] + \frac{(n-1)}{n} \mu^2 - \mu^2 = \frac{1}{n} (E[X^2] - \mu^2) = \frac{1}{n} \sigma_X^2\end{aligned}$$

- The variance of $\hat{\mu}_X$ diminishes as $n \rightarrow \infty$

uncertainty of $\hat{\mu}_X$ $\Rightarrow$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

from Prof. Wu, Shan-Hung's slide

Unbiased Estimator of σ_X^2 (population variance)

- Is $\hat{\sigma}_X^2 = \frac{1}{n} \sum_i (x^{(i)} - \hat{\mu}_X)^2$ an unbiased estimator of σ_X^2 ?

$$\begin{aligned} E_{\mathbb{X}}(\hat{\sigma}_X^2) &= E\left[\frac{1}{n} \sum_i (x^{(i)} - \hat{\mu})^2\right] = E\left[\frac{1}{n} \left(\sum_i x^{(i)2} - 2\sum_i x^{(i)} \hat{\mu} + \sum_i \hat{\mu}^2\right)\right] \\ &= E\left[\frac{1}{n} \left(\sum_i x^{(i)2} - n\hat{\mu}^2\right)\right] = \frac{1}{n} \left(\sum_i E[x^{(i)2}] - nE[\hat{\mu}^2]\right) \\ &= E[X^2] - E[\hat{\mu}^2] = E[(X - \mu)^2 + 2X\mu - \mu^2] - E[\hat{\mu}^2] \\ &= (\sigma^2 + \mu^2) - (\text{Var}[\hat{\mu}] + E[\hat{\mu}^2]) \\ &= \sigma^2 + \mu^2 - \frac{1}{n} \sigma^2 - \mu^2 = \frac{n-1}{n} \sigma^2 \neq \sigma^2 \end{aligned}$$

- What's the unbiased estimator of σ_X^2 ?

$$\hat{\sigma}_X^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_i (x^{(i)} - \hat{\mu}_X)^2 \right) = \frac{1}{n-1} \sum_i (x^{(i)} - \hat{\mu}_X)^2$$


$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

from Prof. Wu, Shan-Hung's slide

The Bernoulli Distribution

- We use the Bernoulli distribution when we have an experiment which can result in one of two outcomes
 - One outcome is labeled “**success**,” and the other outcome is labeled “**failure**”
 - The probability of a success is denoted by p . The probability of a failure is then $1 - p$
- Such a trial is called a **Bernoulli trial** with success probability p

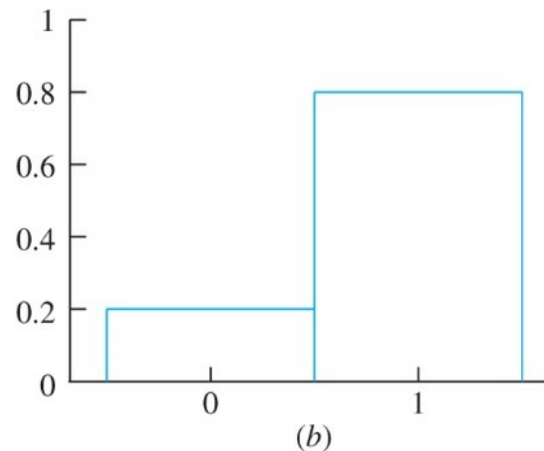
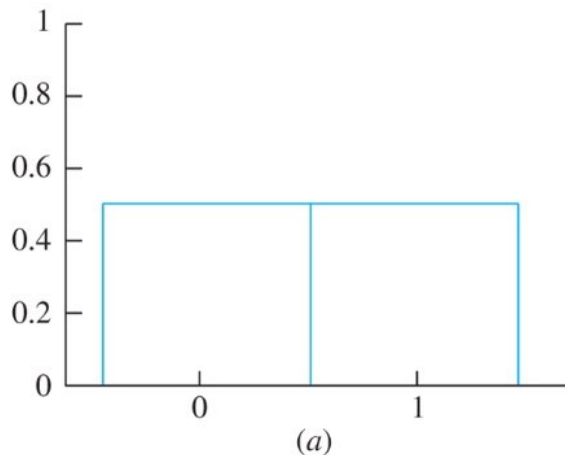
Examples

- The simplest Bernoulli trial is the **toss of a coin**. The two outcomes are heads and tails. If we define heads to be the success outcome, then p is the probability that the coin comes up heads. For a fair coin, $p = \frac{1}{2}$
- Another Bernoulli trial is a **selection of a component from a population of components**, some of which are defective. If we define “success” to be a defective component, then p is the proportion of defective components in the population

$$X \sim \text{Bernoulli}(p)$$

- For any Bernoulli trial, we define a random variable X as follows:
 - If the experiment results in a success, then $X = 1$. Otherwise, $X = 0$. It follows that X is a discrete random variable, with **probability mass function $p(x)$** defined by

$$p_X(x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \end{cases}$$



Mean and Variance of Bernoulli

- If $X \sim \text{Bernoulli}(p)$, then

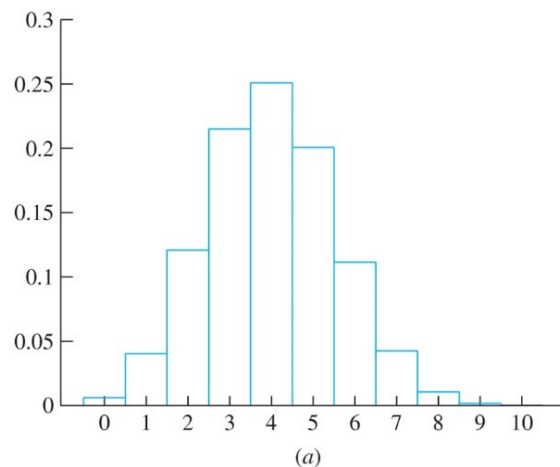
- $\mu_X = \mathbf{E}[X] = \sum x p_X(x) = 1 \cdot p + 0 \cdot (1-p) = p$

- $\mu_{X^2} = \mathbf{E}[X^2] = \sum_x x^2 p_X(x) = 1^2 \cdot p + 0^2 \cdot (1-p) = p$

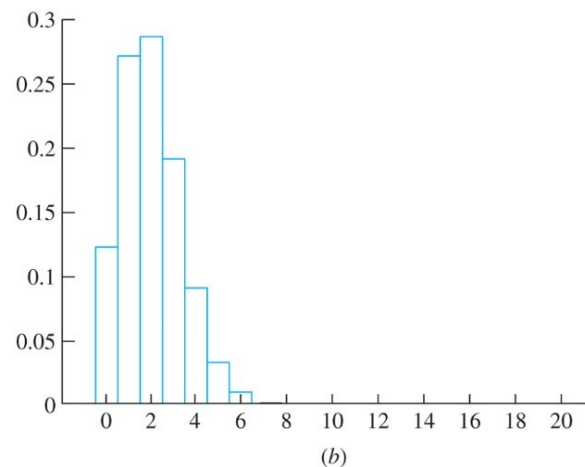
- $\sigma_X^2 = \mu_{X^2} - (\mu_X)^2 = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = p - p^2 = p(1-p)$

The Binomial Distribution

- If a total of n Bernoulli trials are conducted, and The trials are independent
 - Each trial has the same success probability p
 - X is the number of successes in the n trials
- Then X has the **binomial distribution** with parameters n and p , denoted $X \sim \text{Bin}(n, p)$



$\text{Bin}(10, 0.4)$



$\text{Bin}(20, 0.1)$

pmf, Mean and Variance of Binomial (1/2)

- If $X \sim \text{Bin}(n, p)$, the probability mass function of X is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$
$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\sum_{k=0}^n p_X(x) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1$$

– Mean:

$$\begin{aligned} \mu_X &= \sum_x x p_X(x) = \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^n x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^n x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\ &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{n-x} = np \sum_{x'=0}^{n-1} \frac{(n-1)!}{x'!(n-1-x')!} p^{x'} (1-p)^{n-1-x'} \end{aligned}$$

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pmf, Mean and Variance of Binomial (2/2)

- If $X \sim \text{Bin}(n, p)$, the probability mass function of X is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

– Variance:

$$\mathbf{E}[X^2] = \mathbf{E}[X^2 - X] + \mathbf{E}[X] \text{ (to be verified later on!)}$$

$$\begin{aligned} \mathbf{E}[X^2 - X] &= \mathbf{E}[X(X-1)] = \sum_{x=0}^n x(x-1) \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=2}^n x(x-1) \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\ &= n(n-1)p^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)!(n-x)!} p^{x-2} (1-p)^{n-x} = n(n-1)p^2 \end{aligned}$$

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$$\begin{aligned} \sigma_X^2 &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \mathbf{E}[X^2 - X] + \mathbf{E}[X] - (\mathbf{E}[X])^2 \\ &= n(n-1)p^2 + np - n^2 p^2 = np(1-p) \end{aligned}$$

Examples (1/2)

- The probability that a certain kind of component will survive a shock test is $3/4$. Find the probability that exactly 2 of the next 4 components tested survive. (某種元件將在某撞擊測試中通過的機率為 $3/4$ ，請找出在測試的 4 個元件中有兩個通過的機率。)

$$p_X(x) = \mathbf{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

– Solution:

$$p_X(x) = \mathbf{P}(X = 2) = \binom{4}{2} \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 = \left(\frac{4!}{2!2!}\right) \left(\frac{3^2}{4^4}\right) = \frac{27}{128}$$

Examples (2/2)

- The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that (a) at least 10 survive, (b) 3 to 8 survive, and (c) exactly 5 survive? (患者從某種罕見的血液疾病中復原的機率為 0.4，如果有 15 個人已知患有這種疾病，則 (a) 至少 10 人存活；(b) 3 至 8 人存活；(c) 恰好 5 人存活的機率是多少?)

– Solution:

$$\begin{aligned} \text{(a)} \quad p_X(x) &= \mathbf{P}(X \geq 10) = 1 - \mathbf{P}(X < 10) = 1 - \sum_{k=0}^9 \binom{15}{k} 0.4^k (1-0.4)^{15-k} = 1 - 0.9662 \\ &= 0.0338 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad p_X(x) &= \mathbf{P}(3 \leq X \leq 8) = \sum_{k=3}^8 \binom{15}{k} 0.4^k (1-0.4)^{15-k} \\ &= \sum_{k=0}^8 \binom{15}{k} 0.4^k (1-0.4)^{15-k} - \sum_{k=0}^2 \binom{15}{k} 0.4^k (1-0.4)^{15-k} \\ &= 0.9050 - 0.0271 = 0.8779 \end{aligned}$$

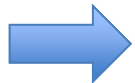
$$\text{(c)} \quad p_X(x) = \mathbf{P}(X = 5) = \binom{15}{5} 0.4^5 (1-0.4)^{15-5} = 0.1859$$

Another Use of the Binomial

- Assume that a finite population contains items of two types, successes and failures, and that a simple random sample is drawn from the population. Then if the sample size is no **more than 5%** of the population, the binomial distribution may be used to model the number of successes
 - Sample items can be therefore assumed to be independent of each other
 - Each sample item is a Bernoulli trial

More on the Binomial

- Assume n independent Bernoulli trials are conducted
- Each trial has probability of success p
- Let $Y_1, \dots, Y_n$ be defined as follows: $Y_i = 1$ if the i -th trial results in success, and $Y_i = 0$ otherwise (Each of the Y_i has the Bernoulli(p) distribution)
- Now, let X represent the number of successes among the n trials. So, $X = Y_1 + \dots + Y_n$



This shows that a binomial random variable can be expressed as a sum of Bernoulli random variables

Estimate of p

- If $X \sim \text{Bin}(n, p)$, then the sample proportion

$$\hat{p} = \frac{\text{number of successes}}{\text{number of trials}} = \frac{X}{n} \left(= \frac{Y_1 + Y_2 + \dots + Y_n}{n} \right)$$

is used to estimate the success probability p

- Note:

- Bias is the difference $\mu_{\hat{p}} - p$

- $\hat{p}$ is unbiased $(\mu_{\hat{p}} - p) = 0$

- The uncertainty in $\hat{p}$ is

$$\sigma_{\hat{p}} = \sigma_{(Y_1 + Y_2 + \dots + Y_n)/n} = \frac{\sqrt{np(1-p)}}{n} = \sqrt{\frac{p(1-p)}{n}}$$

- In practice, when computing $\sigma_{\hat{p}}$, we substitute $\hat{p}$ for p , since p is unknown

The Poisson Distribution

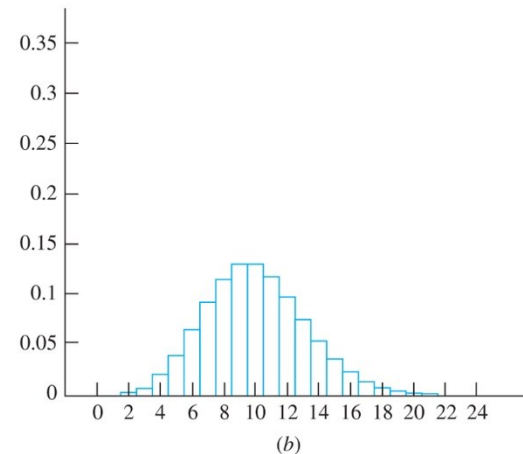
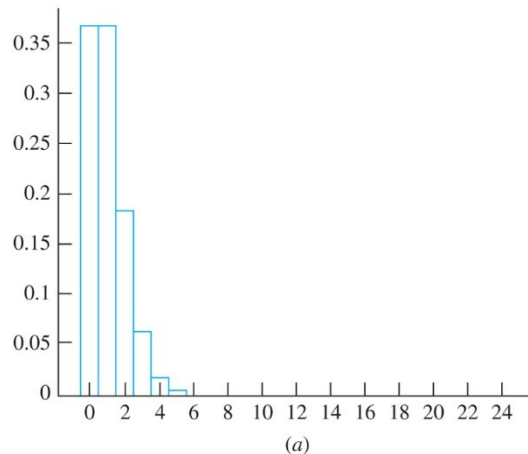
- One way to think of the **Poisson distribution** is as an approximation to the **binomial distribution** when n is large and p is small
- It is the case when n is large and p is small the mass function depends almost entirely on the mean np , very little on the specific values of n and p
- We can therefore approximate the binomial mass function with a quantity $\lambda = np$; this λ is the parameter in the Poisson distribution

pmf, Mean and Variance of Poisson (1/2)

- If $X \sim \text{Poisson}(\lambda)$, the probability mass function of X is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

Probability Histogram



- Note: X must be a discrete random variable and λ must be a positive constant

pmf, Mean and Variance of Poisson (2/2)

- If $X \sim \text{Poisson}(\lambda)$, the probability mass function of X is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

- Mean: $\mu_X = \lambda$

$$\mathbf{E}[X] = \sum_x x p_X(x) = \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = \sum_{x=1}^{\infty} \cancel{x} e^{-\lambda} \frac{\lambda \cdot \lambda^{x-1}}{\cancel{x}(x-1)!} = \lambda \sum_{x'=0}^{\infty} e^{-\lambda} \frac{\lambda^{x'}}{x'!} = \lambda$$

- Variance: $\sigma_X^2 = \lambda$

$$\begin{aligned} \mathbf{E}[X^2] &= \sum_x x^2 p_X(x) = \sum_{x=0}^{\infty} x^2 e^{-\lambda} \frac{\lambda^x}{x!} = \sum_{x=1}^{\infty} \cancel{x} \cdot x e^{-\lambda} \frac{\lambda \cdot \lambda^{x-1}}{\cancel{x}(x-1)!} = \lambda \sum_{x=1}^{\infty} x e^{-\lambda} \frac{\lambda^{x-1}}{(x-1)!} \\ &= \lambda \sum_{x'=0}^{\infty} (x'+1) e^{-\lambda} \frac{\lambda^{x'}}{x'!} = \lambda \left[\left(\sum_{x'=0}^{\infty} x' e^{-\lambda} \frac{\lambda^{x'}}{x'!} \right) + \left(\sum_{x'=0}^{\infty} e^{-\lambda} \frac{\lambda^{x'}}{x'!} \right) \right] = \lambda(\mathbf{E}(X) + 1) = \lambda^2 + \lambda \end{aligned}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Examples (1/3)

- During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond? (在某實驗室的實驗程序中，在一毫秒內通過某個計數器的放射粒子的平均數量為 4，在一個給定的毫秒內有 6 個粒子進入計數器的機率是多少?)

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$\begin{aligned} p_X(x) = \mathbf{P}(X = 6) &= e^{-4} \frac{4^6}{6!} = \sum_{k=0}^6 e^{-k} \frac{4^k}{k!} - \sum_{k=0}^5 e^{-k} \frac{4^k}{k!} \\ &= 0.8893 - 0.7851 = 0.1042 \end{aligned}$$

Examples (2/3)

- Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away? (每天到達某個港口的油輪平均數量為 10 艘。該港口的設施單位每天最多可處理 15 艘油輪，在任意的某一天，油輪必須被迫趕離的機率是多少?)

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$p_X(x) = \mathbf{P}(X > 15) = 1 - \sum_{k=0}^{15} e^{-10} \frac{10^k}{k!} = 1 - 0.9513 = 0.0487$$

Examples (3/3)

- Example 4.18: Particles are suspended in a liquid medium at a concentration of 6 particles per mL. A large volume of the suspension is thoroughly agitated, and then 3 mL are withdrawn. What is the probability that exactly 15 particles are withdrawn?

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

- Solution: $X \sim \text{Poisson}(\lambda)$

The mean number of particles of a 3 mL volume is 18. ($\lambda = 18$)

$$p_X(x) = \mathbf{P}(X = 15) = e^{-18} \frac{18^{15}}{15!} = 0.0786$$

Relation between Binomial and Poisson

- The Poisson PMF with parameter λ is a good approximation for a binomial PMF with parameters n and p , provided that $\lambda = np$, n is very large and p is very small

$$\lim_{n \rightarrow \infty} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \lim_{n \rightarrow \infty} \frac{n!}{(n-k)!k!} p^k (1-p)^{n-k} \quad (\because \lambda = np \Rightarrow p = \frac{\lambda}{n})$$

$$= \lim_{n \rightarrow \infty} \frac{n(n-1)\cdots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \lim_{n \rightarrow \infty} \frac{\lambda^k}{k!} \frac{n(n-1)\cdots(n-k+1)}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \lim_{n \rightarrow \infty} \frac{\lambda^k}{k!} \left(\frac{n}{n}\right) \left(\frac{n-1}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) \left(1 - \frac{\lambda}{n}\right)^{n-k} \quad (\because \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x)$$

$$= \lim_{n \rightarrow \infty} \frac{\lambda^k}{k!} e^{-\lambda}$$

TABLE 4.1 An example of the Poisson approximation to the binomial probability mass function*

x	$P(X = x),$ $X \sim \text{Bin}(10,000, 0.0002)$	$P(Y = x),$ $Y \sim \text{Bin}(5000, 0.0004)$	Poisson Approximation, Poisson (2)
0	0.135308215	0.135281146	0.135335283
1	0.270670565	0.270670559	0.270670566
2	0.270697637	0.270724715	0.270670566
3	0.180465092	0.180483143	0.180447044
4	0.090223521	0.090223516	0.090223522
5	0.036082189	0.036074965	0.036089409
6	0.012023787	0.012017770	0.012029803
7	0.003433993	0.003430901	0.003437087
8	0.000858069	0.000856867	0.000859272
9	0.000190568	0.000190186	0.000190949

*When n is large and p is small, the $\text{Bin}(n, p)$ probability mass function is well approximated by the Poisson (λ) probability mass function (Equation 4.9), with $\lambda = np$. Here $X \sim \text{Bin}(10,000, 0.0002)$ and $Y \sim \text{Bin}(5000, 0.0004)$, so $\lambda = np = 2$, and the Poisson approximation is Poisson(2).

Poisson Distribution to Estimate Rate

- Let λ denote the mean number of events that occur in one unit of time or space. Let X denote the number of events that are observed to occur in t units of time or space
- If $X \sim \text{Poisson}(\lambda t)$, we estimate λ with $\hat{\lambda} = \frac{X}{t}$
- Note:
 - $\hat{\lambda}$ is unbiased ($\mu_{\hat{\lambda}} = \mathbf{E}[\hat{\lambda}] = \mathbf{E}\left[\frac{X}{t}\right] = \frac{1}{t} \mathbf{E}[X] = \frac{1}{t} \cdot \lambda \cdot t = \lambda$)
 - The uncertainty in $\hat{\lambda}$ is $\sigma_{\hat{\lambda}} = \sigma_{\frac{X}{t}} = \sqrt{\frac{1}{t^2} \sigma_X^2} = \sqrt{\frac{1}{t^2} \lambda t} = \sqrt{\frac{\lambda}{t}}$
 - In practice, we substitute $\hat{\lambda}$ for λ , since λ is unknown

Example

- Example 4.24: A 5 mL sample of a suspension is withdrawn, and 47 particles are counted. Estimate the mean number of particles per mL, and find the uncertainty in the estimate?

– Solution:

If $X \sim \text{Poisson}(\lambda t)$, we estimate λ with $\hat{\lambda} = \frac{X}{t}$

$$\hat{\lambda} = \frac{47}{5} = 9.4$$

The uncertainty in the estimate is

$$\begin{aligned}\sigma_{\hat{\lambda}} &= \sqrt{\frac{\lambda}{t}} \\ &= \sqrt{\frac{9.4}{5}} \\ &= 1.4\end{aligned}$$

Some Other Discrete Distributions

- Consider a finite population containing two types of items, which may be called successes and failures
 - A simple random sample is drawn from the population
 - Each item sampled constitutes a Bernoulli trial
 - As each item is selected, the probability of successes in the remaining population decreases or increases, depending on whether the sampled item was a success or a failure
 - For this reason the **trials are not independent**, so the number of successes in the sample does not follow a binomial distribution
- The distribution that properly describes the number of successes is the **hypergeometric distribution**

pmf of Hypergeometric

- Assume a **finite population** contains N items, of which R are classified as successes and $N - R$ are classified as failures
 - Assume that n items are sampled from this population, and let X represent the number of successes in the sample
 - Then X has a hypergeometric distribution with parameters N , R , and n , which can be denoted $X \sim H(N, R, n)$. The probability mass function of X is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{\binom{R}{x} \binom{N-R}{n-x}}{\binom{N}{n}}, & \text{if } \max(0, R+n-N) \leq x \leq \min(n, R) \\ 0, & \text{otherwise} \end{cases}$$

Binomial PMF: sampling with replacement of each item to ensure independence between trials

Example

- Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and to reject the lot if a defective is found. What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot? (每批貨有 40 個元件，如果含有 3 個或更多個瑕疵品，則該批貨被認為是不可接受的，抽樣的程序是隨機選擇 5 個元件，如果發現有瑕疵，則拒絕該批貨。如果一批貨中有 3 個瑕疵，那麼在樣本中發現正好有 1 個瑕疵的機率是多少?)

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{\binom{R}{x} \binom{N-R}{n-x}}{\binom{N}{n}}, & \text{if } \max(0, R+n-N) \leq x \leq \min(n, R) \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$p_X(x) = \mathbf{P}(X = 1) = \frac{\binom{3}{1} \binom{37}{4}}{\binom{40}{5}} = 0.3011, \quad 0 \leq x \leq 3$$

How about $\mathbf{P}(X \leq 1)$?

Mean and Variance of Hypergeometric

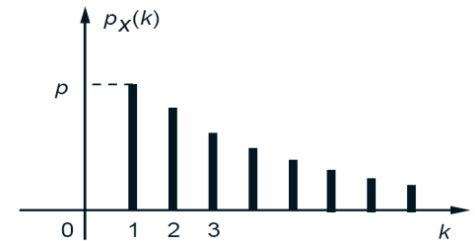
- If $X \sim H(N, R, n)$, then
 - Mean of X : $\mu_X = \frac{nR}{N}$
 - Variance of X : $\sigma_X^2 = n \left(\frac{R}{N} \right) \left(1 - \frac{R}{N} \right) \left(\frac{N-n}{N-1} \right)$

Geometric Distribution

- Assume that a sequence of independent Bernoulli trials is conducted, each with the same probability of success, p
- Let X represent the number of trials up to and including the first success
 - Then X is a discrete random variable, which is said to have the **geometric distribution** with parameter p .
 - We write $X \sim \text{Geom}(p)$.

pmf, Mean and Variance of Geometric

- If $X \sim \text{Geom}(p)$, then



- The pmf of X is $p_X(x) = \mathbf{P}(X = x) = \begin{cases} p(1-p)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$

- Mean of X is $\mu_X = \frac{1}{p}$

- Variance of X is $\sigma_X^2 = \frac{1-p}{p^2}$

Mean and Variance of the **Geometric**

- Consider a **geometric** random variable with a PMF

$$p_X(x) = (1-p)^{x-1} p, \quad x = 1, 2, \dots,$$

$$\mathbf{E}[X] = \sum_x x p_X(x) = \sum_{x=0}^{\infty} x (1-p)^{x-1} p = p \sum_{x=1}^{\infty} x q^{x-1} \quad (\text{let } q = 1-p < 1)$$

$$= p \frac{d\left(\sum_{x=0}^{\infty} q^x\right)}{dq} = p \frac{d\left(\frac{1}{1-q}\right)}{dq} = p \frac{1}{(1-q)^2} = \frac{1}{p}$$

Note that: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$, for $|r| < 1$

$$\mathbf{E}[X^2] = \mathbf{E}[X^2 - X] + \mathbf{E}[X] \quad (\text{to be verified later on!})$$

$$\mathbf{E}[X^2 - X] = \mathbf{E}[X(X-1)] = \sum_{x=0}^{\infty} x(x-1)(1-p)^{x-1} p = pq \sum_{x=2}^{\infty} x(x-1)q^{x-2} \quad (\text{let } q = 1-p < 1)$$

$$= pq \sum_{x=2}^{\infty} x(x-1)q^{x-2} = pq \frac{d^2\left(\frac{1}{1-q}\right)}{d^2q} = pq \frac{2}{(1-q)^3} = \frac{2(1-p)}{p^2}$$

$$\begin{aligned} \text{var}(X) &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \mathbf{E}[X^2 - X] + \mathbf{E}[X] - (\mathbf{E}[X])^2 \\ &= \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{(1-p)}{p^2} \end{aligned}$$

Example

- At a “busy time,” a telephone exchange is very near capacity, so callers have difficulty placing their calls. It may be of interest to know the number of attempts necessary in order to make a connection. Suppose that we let $p = 0.05$ be the necessary in order to make a connection during a busy time. We are interested in knowing the probability that 5 attempts are necessary for a successful call. (在「繁忙時間」中電話交換機非常接近滿載，因此打電話者要打通電話可能會有困難。我們想知道成功通話所需要嘗試的次數。假設我們令 $p = 0.05$ 是繁忙時間連接成功的紀律，則成功通話需要 5 次嘗試的機率為何?)

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} p(1-p)^{x-1}, & x = 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$p_X(x) = \mathbf{P}(X = 5) = (1 - 0.05)^{5-1} (0.05) = (0.95)^4 (0.05) = 0.041$$

Negative Binomial Distribution

- The negative binomial distribution is an extension of the geometric distribution. Let r be a positive integer.
- Assume that independent Bernoulli trials, each with success probability p , are conducted, and let X denote the number of trials up to and including the r -th success
 - Then X has the **negative binomial distribution** with parameters r and p . We write $X \sim \text{NB}(r, p)$
- Note: If $X \sim \text{NB}(r, p)$, then $X = Y_1 + \dots + Y_r$ where $Y_1, \dots, Y_r$ are independent random variables, each with $\text{Geom}(p)$ distribution

$$\underbrace{0, \dots, 1}_{Y_1} \underbrace{0, \dots, 1}_{Y_2} \underbrace{0, \dots, 1}_{Y_r} \dots, 0, \dots, 1 \quad \text{x trials in total}$$

pmf, Mean and Variance of Negative Binomial

- If $X \sim \text{NB}(r, p)$, then

- The pmf of X is $p_X(x) = \mathbf{P}(X = x) = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x = r, r+1, \dots \\ 0, & \text{otherwise} \end{cases}$

- Mean of X is $\mu_X = \frac{r}{p}$

- Variance of X is $\sigma_X^2 = \frac{r(1-p)}{p^2}$

Example

- In an NBA championship series, the team that wins four games out of seven is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B .
 - (a) What is the probability that team A will win the series in 6 games?
 - (b) What is the probability that team A will win the series?
 - (c) If team A and B were facing each other in a regional playoff series, which is decided by winning three out of five games, what is the probability that team A would win the series?

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x = r, r+1, \dots \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$(a) \quad p_X(x) = \mathbf{P}(X = 6) = \binom{6-1}{4-1} (0.55)^4 (1-0.55)^{6-4} = 0.1853$$

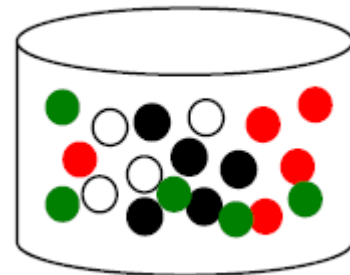
$$(b) \quad p_X(x) = \mathbf{P}(4 \leq X \leq 7) = \sum_{x=4}^7 p_X(x) = \mathbf{P}(X = 4) + \mathbf{P}(X = 5) + \mathbf{P}(X = 6) + \mathbf{P}(X = 7)$$

$$= 0.0915 + 0.1647 + 0.1853 + 0.1668 = 0.6083$$

$$(c) \quad p_X(x) = \mathbf{P}(3 \leq X \leq 5) = \sum_{x=3}^5 p_X(x) = \mathbf{P}(X = 3) + \mathbf{P}(X = 4) + \mathbf{P}(X = 5)$$

$$= 0.1664 + 0.2246 + 0.2021 = 0.5931$$

Multinomial Distribution



- Bernoulli trial is a process that results in one of two possible outcomes. A generalization of the Bernoulli trial is the **multinomial trial**, which is a process that can result in any of k outcomes, where $k \geq 2$. We denote the probabilities of the k outcomes by $p_1, \dots, p_k$ ($p_1 + \dots + p_k = 1$)
- Now assume that n independent multinomial trials are conducted each with k possible outcomes and with the same probabilities $p_1, \dots, p_k$. Number the outcomes 1, 2, ..., k . For each outcome i , let X_i denote the number of trials that result in that outcome. Then $X_1, \dots, X_k$ are discrete random variables. The collection $X_1, \dots, X_k$ is said to have the **multinomial distribution** with parameters $n, p_1, \dots, p_k$. We write $X_1, \dots, X_k \sim \text{MN}(n, p_1, \dots, p_k)$

pmf of Multinomial

- If $X_1, \dots, X_k \sim \text{MN}(n, p_1, \dots, p_k)$, then the pmf of $X_1, \dots, X_k$ is

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}, & x_i = 0, 1, 2, \dots, n; \text{ and } \sum x_i = n \\ 0, & \text{otherwise} \end{cases}$$

Can be viewed as a joint probability mass function of $X_1, \dots, X_k$

- Note that if $X_1, \dots, X_k \sim \text{MN}(n, p_1, \dots, p_k)$, then for each i , $X_i \sim \text{Bin}(n, p_i)$

Example

- For a certain airport containing three runways it is known that in the ideal setting the following are the probabilities that the individual runways are accessed by a randomly arriving commercial jet:

Runway 1: $p_1 = 2/9$, Runway 2: $p_2 = 1/6$, Runway 3: $p_3 = 11/18$

What is the probability that 6 randomly arriving airplanes are distribution in the following fashion?

Runway 1: 2 airplanes,

Runway 2: 1 airplane,

Runway 3: 3 airplanes

$$p_X(x) = \mathbf{P}(X = x) = \begin{cases} \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}, & x_i = 0, 1, 2, \dots, n; \text{ and } \sum x_i = n \\ 0, & \text{otherwise} \end{cases}$$

– Solution:

$$p_X(x) = \binom{6}{2,1,3} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3 = \left(\frac{6!}{2! \cdot 1! \cdot 3!}\right) \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right)^1 \left(\frac{11}{18}\right)^3 = 0.1127$$